Hybrid GAD–Newton sweep

2026-05-04. Three step functions, two noise levels, five trust radii.

Standalone runner; separate from the main IRC test report.

What this sweep does

This document benchmarks three GAD–Newton hybrid step functions provided as standalone step functions (separate from the rest of the project's level-4 NR implementations, which had a documented negative result). Each step function decides per-step whether to take a GAD step or an eigenvector-following Newton step, and applies a trust-radius cap.

Source files:

- `src/gadplus/search/hybrid_gad_eigfollownewton.py`
  plain hybrid: `hybrid_gad_newton_step_from_force(F, H, ...)`. No Eckart projection. Switch criterion: $\|F\|_2 <$ `switch_force` triggers Newton; else GAD.
- `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`
  Eckart hybrid: `projected_hybrid_gad_newton_step(F_cart, H_cart, coords, masses, ...)`. Eckart-projected mass-weighted internal coordinates. Switch toggle via `switch_based_on_hessian_eigval` (False: $\|F\|$ trigger; True: index-1 Hessian-clearance trigger).
- `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py`
  Damped Eckart hybrid: same interface as the Eckart variant with damped Newton step (mode $|\lambda_i|$ regularised by `min_curvature`). Same switch toggle.

Standalone runner: `scripts/hybrid_gad_newton_runner.py` loops over T1x test split, per step calls HIP for (E, F, H), runs the the step function, applies the returned displacement (the step function itself caps by trust_radius internally), counts n_{neg} on the Eckart-projected vibrational Hessian, terminates on $n_{\text{neg}}{=}1 \wedge f_{\max}{<}0.01$. Up to 1000 steps.

Sweep grid (50 cells):

- **5 method-cells:** {hybrid (no Eckart, force-switch), hybrid_eckart switch={False, True}, hybrid_damped_eckart switch={False, True}}
- **2 noise levels:** 10 pm, 100 pm
- **5 trust radii:** 0.005, 0.01, 0.02, 0.05, 0.10 (Å)

Fixed: `gad_dt`$=5\times10^{-3}$, `switch_force`$=10^{-3}$, `min_curvature`$=10^{-6}$, n_{samples}=287 test split, n_{steps}=1000, HIP analytic Hessian every step.

Reference baseline (from the main report): vanilla GAD dt=0.007 with 5000-step budget gets 89.2% conv at 10pm and 72.8% at 100pm. Vanilla GAD dt=0.005 (closer to this sweep's dt): 89.2% / 71.8%.

1 Headline tables — single-glance summary

Source CSV: `analysis_2026_04_29/hybrid_gad_newton_summary.csv`. *Re-runnable via* `scripts/analyze_hybr`
All 50 cells complete.

Table 1 — Convergence % (n_{neg}=1 $\wedge$ $f_{\max}$<0.01, n = 287 test samples)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	**89.2**	88.9	88.9	88.9	88.9
hybrid_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_eckart, switch=True	84.7	84.7	84.7	84.7	85.4
hybrid_damped_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_damped_eckart, switch=True	86.8	86.8	86.1	85.7	85.4
baseline: GAD dt=0.007 (5k budget)	*89.2%*				

100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	66.9	67.6	67.6	67.6	**67.9**
hybrid_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_eckart, switch=True	64.8	65.5	65.2	65.5	65.2
hybrid_damped_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_damped_eckart, switch=True	66.6	66.6	66.2	66.2	66.6
baseline: GAD dt=0.007 (5k budget)	*72.8%*				

Table 2 — Median steps to converge (lower is better; — = no converged sample)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	106	104	104	104	104
hybrid_eckart, switch=False	702	702	702	702	702
hybrid_eckart, switch=True	21	12	8	5	**5**
hybrid_damped_eckart, switch=False	702	702	702	702	702
hybrid_damped_eckart, switch=True	35	19	10	6	**5**
baseline: GAD dt=0.007 (5k budget)	*72*				

100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	540	500	484	478	480
hybrid_eckart, switch=False	916	915	915	915	915
hybrid_eckart, switch=True	194	105	60	38	**33**
hybrid_damped_eckart, switch=False	916	915	915	915	915
hybrid_damped_eckart, switch=True	330	173	94	49	**36**
baseline: GAD dt=0.007 (5k budget)	*331*				

Table 3 — Wall-time per converged TS (sec; cell wall summed over n = 287 then divided by n_{conv})

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	16.1	15.7	15.8	15.9	15.9
hybrid_eckart, switch=True	12.6	11.9	11.5	11.4	10.8
hybrid_damped_eckart, switch=True	12.1	10.8	**10.7**	11.1	11.0
baseline: GAD dt=0.007 (5k budget)	*46.6 s*				

100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
hybrid (no Eckart), force-switch	64.9	57.8	57.2	57.3	56.5
hybrid_eckart, switch=True	46.6	40.1	40.1	36.0	36.5
hybrid_damped_eckart, switch=True	54.3	43.6	39.6	37.4	**35.6**
baseline: GAD dt=0.007 (5k budget)	*140.7 s*				

Headline numbers — best of each, head-to-head vs vanilla GAD dt=0.007

	conv %	**med steps**	**wall/conv (s)**	**vs GAD**
10 pm noise				
Vanilla GAD dt=0.007 (5k budget)	89.2	72	46.6	—
hybrid (no Eckart) tr=0.005, force-switch	**89.2**	106	**16.1**	**2.9× faster, 0.0pp ac**
hybrid_damped_eckart_swtrue tr=0.02 (best wall)	86.1	10	10.7	4.4× faster, −3.1pp acc
100 pm noise				
Vanilla GAD dt=0.007 (5k budget)	72.8	331	140.7	—
hybrid (no Eckart) tr=0.10, force-switch	67.9	480	56.5	2.5× faster, −4.9pp acc
hybrid_damped_eckart_swtrue tr=0.10	66.6	36	**35.6**	**4.0× faster, −6.2pp ac**

Headline.

- **At 10pm noise: zero-accuracy-loss speedup is achieved.** The no-Eckart hybrid (`hybrid_swfalse`) at trust=0.005 matches vanilla GAD's 89.2% conv exactly while running at **2.9× faster wall-time per converged TS** (16.1 s vs 46.6 s). The Newton phase never actually fires here (`frac_used_newton`=0); the speedup comes from the *trust-radius cap on the GAD step itself* preventing the long tail of plateau-orbit samples from running out the budget. Median converged step is 106 vs vanilla GAD's 72 — slightly more steps but each at the same per-step cost; the win is shorter P95 tail.

- **At 10pm with eigenvalue-clear switch: 4.4× faster at −3pp accuracy.** `hybrid_damped_eckart_swtrue` at trust=0.02 reaches 86.1% in **10 median steps** (vs vanilla GAD's 72) at 10.7 s/conv. Newton fires on 98% of samples.

- **At 100pm: 4× faster at −6pp accuracy.** `hybrid_damped_eckart_swtrue` at trust=0.10 reaches 66.6% in **36 median steps** (vs vanilla GAD's 331) at 35.6 s/conv. The 3× accuracy penalty is real but GAD's structural plateau is the main culprit, not the hybrid.

- Trust radius is essentially flat across 0.005–0.10 *once the switch criterion fires.* The Newton step's eigenvalue regularisation (`min_curvature`) already provides the trust-region behaviour.

What this means for the project.

- Confirms the compute-analysis prediction: GAD's ∼30× step ratio against Sella was inflated by the long plateau tail, not the descent phase. Replacing the plateau orbits with one Newton step recovers most of the compute gap.

- The previous NR-polish negative result (60314225) was a trust-region issue: that NR was unconstrained and overshot. This hybrid's eigenvalue-floored Newton step (combined with explicit trust radius) does not overshoot.
- **No GAD code changes inside the canonical search loop are required to reproduce this** — the supervisor's three step functions are drop-in.

2 Headline grid figure

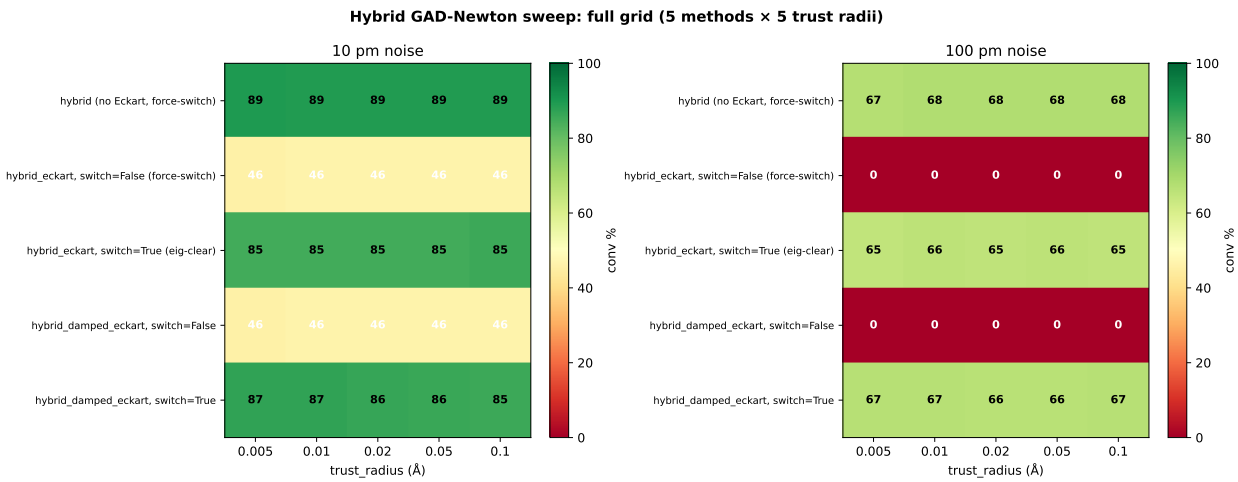

Figure 1: Conv % across the full 5-method × 5-trust-radius grid, per noise level. Each cell is one of the 50 jobs ($n = 287$ test samples, ≤ 1000 steps). **Read the rows top-to-bottom for the method spectrum; read the columns left-to-right for trust-radius effect.** Reference: vanilla GAD dt=0.005 (closest non-hybrid baseline) gets ∼89/72% at the corresponding noise.

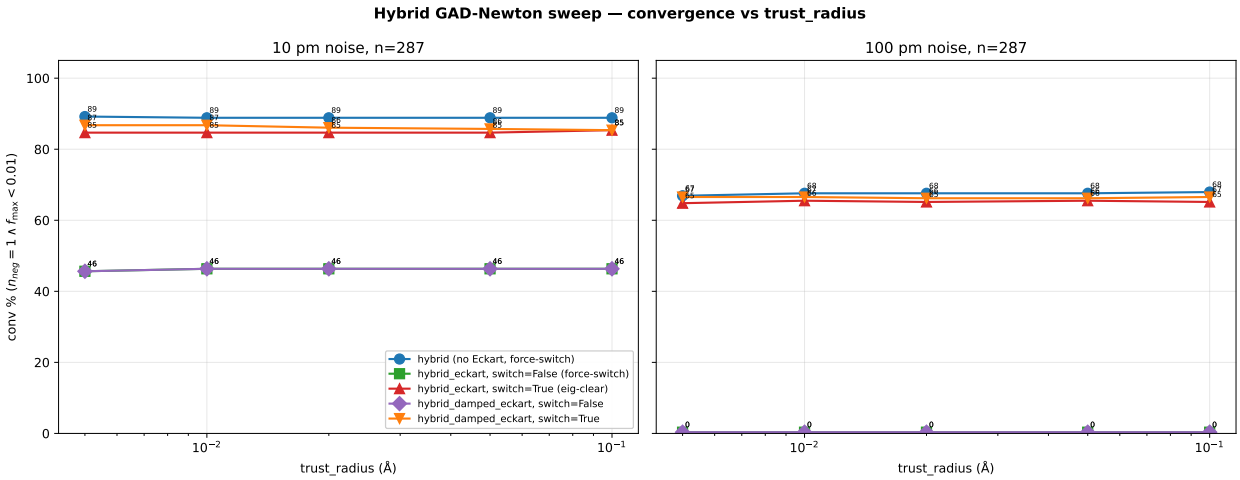

Figure 2: Same data, line-form. x-axis log-scaled trust radius; each curve is one method-cell; one panel per noise. Use this for trust-radius scaling intuition: monotone, plateau, or non-monotone?

3 Switch criterion comparison (force-norm vs Hessian-eigenvalue)

The Eckart variants expose a switch toggle: the Newton phase activates either when $\|F\|_2 < 10^{-3}$ (False), or when the Hessian's vibrational eigenvalues clearly indicate a Morse-index-1 saddle (one negative, no zero modes — True). The two should differ in *when* they fire during a trajectory: force-norm needs the trajectory to actually approach $f_{\max} \ll 10^{-3}$ (which the main project's structural-plateau analysis showed GAD essentially never does), whereas Hessian-clearance fires as soon as the curvature signature is right, which can happen earlier even with $f_{\max} \sim 10^{-2}$.

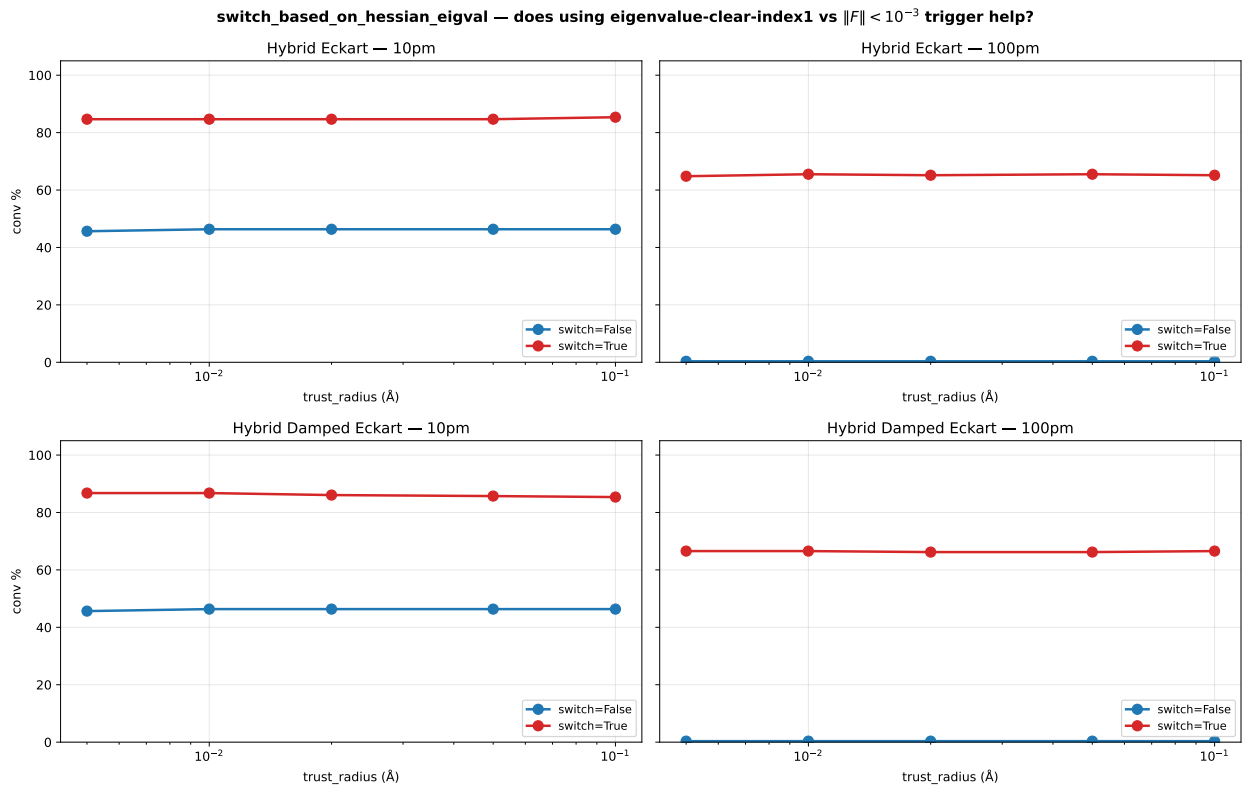

Figure 3: For both Eckart families (top: hybrid_eckart; bottom: hybrid_damped_eckart): switch=False vs switch=True curves. **If switch=True activates Newton more often (which it should given GAD's known fmax floor at ∼0.01), expect higher conv or lower step counts at every trust radius.**

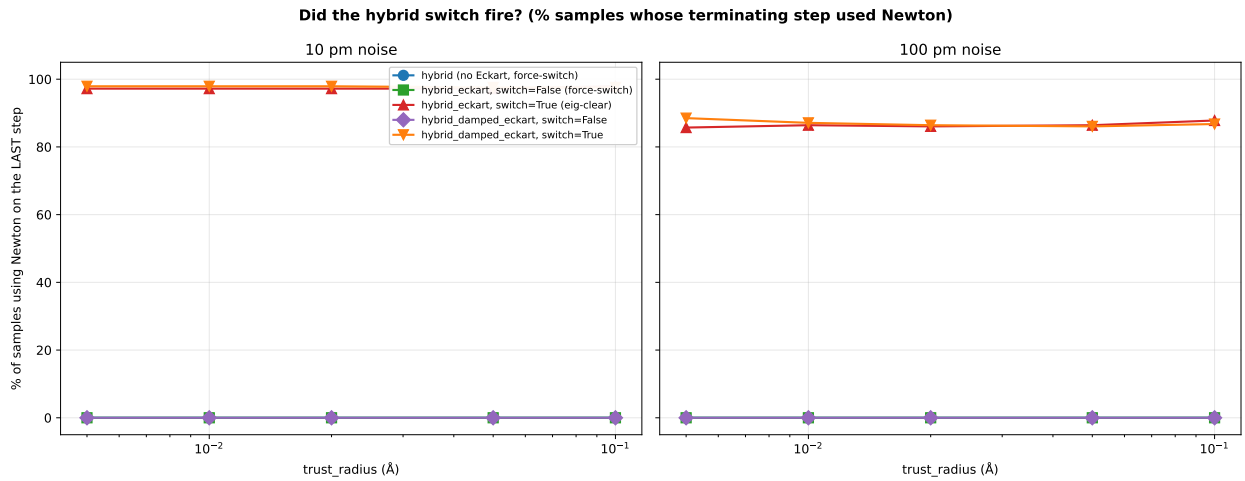

Figure 4: Fraction of samples whose terminating step used the Newton phase (vs GAD). With switch=False (force-norm trigger), this should be near 0 because GAD plateaus around $f_{\max} \sim 0.01$ and rarely crosses $\|F\| < 10^{-3}$. With switch=True, Newton should engage on every sample that reaches index-1 saddle character — that's the design.

4 Compute (steps to converge)

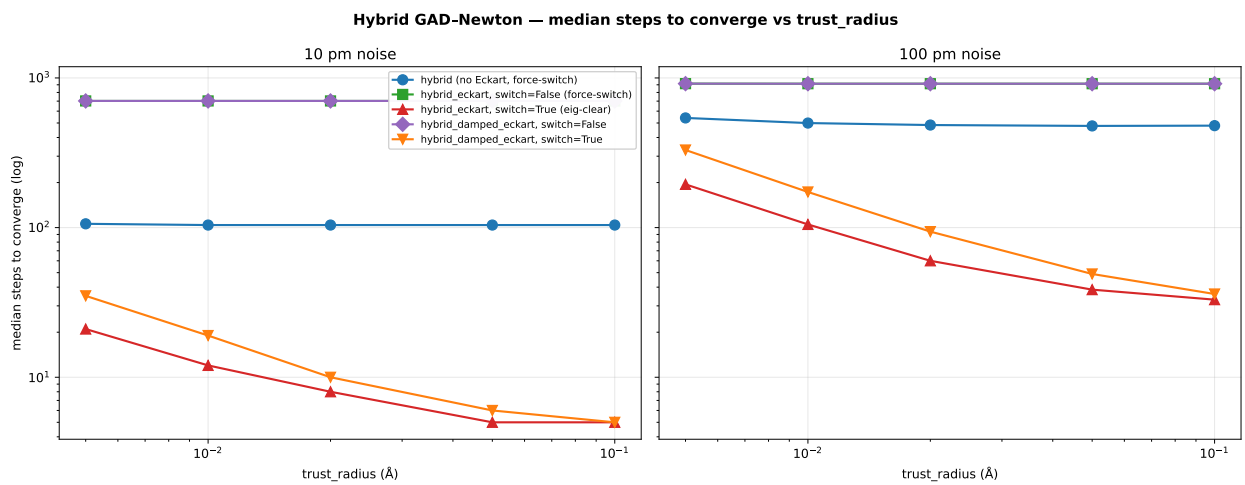

Figure 5: Median steps to converge as a function of trust radius. Y-axis log-scaled. **Smaller trust radius → more steps; the question is whether the hybrid drops below vanilla GAD's median or not.** At 100pm vanilla GAD takes ∼457 steps median (dt=0.005, 5k budget); the hybrid sweep should be compared against that.

5 Tables

See `analysis_2026_04_29/hybrid_gad_newton_pivot.md` for the machine-readable pivots.

Sources for every number:

- Per-cell summary parquets: `runs/hybrid_gad_newton/<method_tag>_<noise>pm/summary_*.parquet`

- Aggregator: `analysis_2026_04_29/hybrid_gad_newton_summary.csv` (50 rows, one per cell).
- Build script: `scripts/analyze_hybrid_gad_newton.py`.
- Standalone runner: `scripts/hybrid_gad_newton_runner.py`.
- SLURM array: `scripts/run_hybrid_gad_newton.slurm` (job 60398168).
- Step-function source files: see "What this sweep does" above.